BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: KT-XII-21-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Code D

Note: Q.1 Write the correct option i.e. A, B, C or D in the empty box provided opposite to each part. No marks will be awarded for cutting, erasing or over writing.

Qi. Which one bacterium is Aerobic .....

(A) Azobactor
(B) Clostridium
(C) Azotobacter
(D) None of these

Qii. Which one is stop codons?

(A) CAU
(B) UAG
(C) UAA
(D) Both B and C

Qiii. During development, nervous system and the skin epidermis is developed by .....

(A) Endoderm
(B) Mesoderm
(C) Ectoderm
(D) Endometrium

Qiv. Which one is the salient feature of Darwin-Wallace Theory?

(A) Variation
(B) Over production
(C) Struggle for existence
(D) All of these

Qv. Fungiform Papillae is of .....shape.

(A) Cone
(B) Mushroom
(C) Flat
(D) None of these

Qvi. Effectors may be.....

(A) Bone
(B) Gland
(C) Muscle
(D) Both A and B

Qvii. Inheritance of skin colour in man is controlled by eight pairs of genes which are ……

(A) Linked
(B) Codominent
(C) Multiple alleles
(D) Assorting Independently

Qviii. A skinny pouch covering the testes is called ……

(A) Vasa efferentia
(B) Vas deferens
(C) Semen
(D) Scrotum

Qix. A sarcomere is the region of myofibril between two successive ……line.

(A) M
(B) I
(C) H
(D) Z

Qx. Cellular rearrangement and migration occurs during……staige.

(A) Morulla
(B) Blastilla
(C) Ampula
(D) Gastrulation

Qxi. Carbon dioxide is mainly transported as ……

(A) Bicarbonate ions
(B) Carboxyl haemoglobin
(C) Dissolved CO_2 in plasma
(D) Combined with NH_3

Qxii. Thermus aquaticus is the source of……

(A) Primase Enzyme
(B) Taq Polymerase
(C) Vent Polymerase
(D) Both A and B

Qxiii. The hormone which is used as most potent stimulator of bronchial dilation and increased blood flow to heart is……

(A) Norepinephrine
(B) Epinephrine
(C) Thyroxine
(D) Calcitonin

Qxiv. The act of emptying the bladder is called……

(A) Sphincters
(B) Micturition
(C) Urectelic
(D) None of these

Qxv. ……nerves carry impulses from CNS to the effectors.

(A) Sensory Nerves
(B) Motor Nerves
(C) Mixed Nerves
(D) Both B & C

Qxvi. What is the total number of bones in the cranium of human?

(A) 02
(B) 05
(C) 06
(D) 08

Qxvii. Pain receptors are called.....

(A) Pacinians
(B) Nociceptors
(C) Tactile receptor
(D) Mechanoreceptor

Qxviii. Xo-XX type of sex determination mechanism is present in.....

(A) Human
(B) Drosophila
(C) Grasshopper
(D) Moth

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

QQ.2. Attempt any TEN parts. All parts carry equal marks.

i. Discuss Osmoregulatory function of Human Kidney.

ii. Describe the causes of Female Infertility.

iii. What do you know about German Measles?

iv. Differentiate between Identical and Fraternal Twins.

v. Discuss genetic basis of ABO blood group system.

vi. What is Olfaction?

vii. Differentiate between sex-limited and sex-influenced traits.

viii. What is biogas? Also explain the uses of biogas.

ix. Discuss Bombay Phenotype.

x. Draw a label diagram of the structure of Nephron.

xi. Differentiate between primary succession and secondary succession.

xii. Discuss Sinusitis.

xiii. Discuss four applications of PCR.

Section "C"

Maximum Marks: 27

Note: Note: Attempt any THREE questions. All questions carry equal marks.

QQ.3. Discuss in detail the patterns of Sex determination.

QQ.4. Write a detail note on nitrogen cycle.

QQ.5. Write a short notes on the following;

i. Foetal development. The second trimesters.

ii. Discuss dispersive model of DNA.

iii. Define monosomy, trisomy and tetrasomy.

QQ.6. Discuss in detail how Mendel performed dihybrid cross.